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Tzu-Hsien Lee

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Research Interests

Robot Learning for Real-World Manipulation; Data-Efficient Imitation Learning; Vision-Based Perception and Action; Learning-Based Robot Positioning

Education

M.S. in Robotics, University of Minnesota, Twin Cities 09/2023 - Present

- Advisor: Prof. Karthik Desingh
- Courses: Robot Vision, Computer Vision, Robotics, Deep Learning for Robotics, Optimal Estimation

B.S. in Mechanical Engineering, National Yang Ming Chiao Tung University 09/2018 - 06/2022

- Courses: Robotics, Object Oriented Programming, Microcomputer System and Lab, Nonlinear Control

Publications

- **T. Lee**, F. Mahmudova, K. Desingh. "Learning Category-level Last-meter Navigation from RGB Demonstrations of a Single-instance." Accepted, *IEEE Robotics and Automation Letters (RA-L)*, 2026. **[Project Page]**

Presentations

- **Poster Presentation**. "Learning Object-Centric Local Navigation from RGB Demonstrations." Midwest Robotics Workshop (MWRW), TTIC, Chicago, IL, June 2025.
- **Student Lecture**. "RGB-D Networks." CSCI 5980: Deep Learning for Robot Manipulation, University of Minnesota, Minneapolis, MN, November 2024.

Experience

Graduate Researcher, Robotics: Perception and Manipulation Lab 09/2024 - Present

- First-authored RA-L paper (accepted, 2026) on category-level last-meter navigation for mobile manipulation, achieving 96.94% success by training goal-conditioned imitation learning policies on multi-view RGB demonstrations from a single object instance, generalizing across the category on Boston Dynamics Spot.
- Designed the policy architecture with a frozen DINOv2 vision encoder, language-driven SAM2 segmentation, and a score-matrix decoder that captures spatial correlation between current and goal observations for action prediction.
- Accelerated real-world robot learning research by designing and implementing SpotStack, a modular software library for perception integration, data collection, and policy deployment on Boston Dynamics Spot.
- Built semantic topological mapping pipelines with vision-language foundation models for spatially-aware navigation and downstream manipulation on Boston Dynamics Spot.
- Exploring one-shot generalization for eye-in-hand manipulation by studying in-context imitation learning from a single demonstration using wrist-mounted visual observations.

Robotics Course Developer, CSCI 5551: Robotics, University of Minnesota 01/2026 - 05/2026

- Designed and engineered a 10-stage real-robot manipulation curriculum by integrating a 6-DoF Lite6 robotic arm and a ZED 2i stereo camera for perception-to-manipulation learning.
- Enabled rapid development of vision-based manipulation pipelines by building reusable software infrastructure, including AprilTag-based camera-to-robot calibration, grasp primitives, and camera wrapper APIs.
- Implemented reference solutions for zero-shot target selection and 6D object pose estimation using SAM-based segmentation and oriented bounding box fitting on point clouds.
- Established progressive checkpoints from single-object grasping to multi-object sequential stacking with increasing perception difficulty.
- Improved usability and reproducibility by authoring comprehensive 20-page technical documentation and starter templates.

Teaching Assistant, CSCI 5561: Computer Vision, University of Minnesota 09/2025 - 01/2026

- Designed and maintained an automated grading system for coding assignments, enabling scalable and consistent evaluation of student submissions.
- Held weekly office hours and provided technical guidance on computer vision algorithms and deep learning methods, including feature extraction, image registration, scene recognition, and CNN implementation.

- Contributed to midterm exam design and grading, and managed overall course grading logistics.

Robotics Intern, Industrial Technology Research Institute 11/2022 - 02/2023

- Established an autonomous grinding production line for irregularly shaped water faucets by designing force-adaptive trajectories using a robot arm and a force feedback sensor.
- Improved on-site calibration efficiency by developing a coordinate transformation algorithm for adaptive trajectory tuning.

Software Lead, NYCU Autonomous Underwater Vehicle (AUV) Team 02/2021 - 09/2022

- Spearheaded the design and implementation of the software architecture, control system, and navigation, winning Silver Medal at the 2021 Intelligent Innovation Contest and qualifying for SAUVC 2022 (93 teams, 19 countries).
- Established a modular high-level/low-level robotics system by using an embedded system (STM32) for low-level attitude and motion control, and deploying ROS on Raspberry Pi and Jetson Nano for task planning and navigation.
- Reduced high-frequency noise in attitude estimation by 10% through implementing a gradient descent-based Madgwick filter on embedded hardware.
- Developed a real-time obstacle-aware navigation system by employing A* algorithm and sonar-based localization.

Selected Projects

One-Shot Imitation Learning via Visual Servoing 11/2024 – 12/2024

- Achieved one-shot manipulation generalization by implementing a two-stage imitation learning framework combining object-centric segmentation and learned visual servoing using PyTorch and ROS.
- Enabled closed-loop end-effector control from visual observations by training a Siamese CNN servoing policy on features extracted from a FiLM-conditioned U-Net segmentation model.
- Demonstrated generalization to unseen objects by evaluating the policy in ROS Gazebo using a Kinova robotic arm.

Visual SLAM in Dynamic Environments 04/2024 - 05/2024

- Achieved 0.005 rad rotation and 0.36 m translation error by developing a visual SLAM pipeline using ORB feature matching, FLANN search, and RANSAC-based pose refinement.
- Improved robustness in dynamic scenes by filtering moving objects using scene flow and KL divergence before pose estimation.

LiDAR-guided Autonomous Mobile Robot in Unstructured Environments 04/2024 - 05/2024

- Achieved autonomous exploration and package delivery by developing a navigation system using LiDAR-based feature extraction for perception and the Bug algorithm for path planning on a real-world TurtleBot.

Object Assembly Using Visual Servoing 11/2023 - 12/2023

- Achieved accurate object assembly by implementing a relative pose registration pipeline using visual features and the Iterative Closest Point (ICP) algorithm in CoppeliaSim with two camera-equipped robotic manipulators.

Technical Skills

Languages: C, C++, Python, MATLAB, JavaScript
Software & Tools: PyTorch, ROS, Gazebo, OpenCV, Qt, Git, Docker, SolidWorks
Robotics Platforms & Systems: Boston Dynamics Spot, FANUC robots, STM32, Raspberry Pi, Jetson Nano

Leadership & Service

Appointed Treasurer, NYCU Student Association 01/2021 - 09/2021

- Allocated and managed $30,000 budget, approved and recorded all financial transactions, and provided monthly financial reports to the Student Council.